

Speaker Identification and Voice-Activity Detection in Continuous Meeting Audio

T-03, Phase 2 Report

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1 Narrow Project Question and Scope

The narrowed question is:

In three continuous 300-second excerpts from the AMI Meeting Corpus, how can a simple inference-only pipeline combining pretrained Silero VAD and ECAPA-TDNN detect speech and identify the active speaker among 12 enrolled speakers?

The task uses closed-set speaker identification: every single-speaker interval eligible for speaker-identification accuracy belongs to one of the 12 enrolled identities. This differs from open-set verification, which must accept or reject a claimed identity and handle voices not present during enrollment. The proposed baseline has no unknown-speaker rejection rule, so open-set claims are outside the project scope. Overlap is retained in the continuous recordings as a realistic failure condition, but the baseline is not a full diarisation or speech-separation system.

2 Dataset

2.1 Source and selected subset

The AMI Meeting Corpus is a public multimodal corpus containing approximately 100 hours of recorded meetings, with synchronised audio and several forms of annotation [1, 2]. It is appropriate for T-03 because its natural multi-party recordings include speech, silence, speaker changes, and overlapping speech from multiple speakers. The present project uses the mono **Mix-Headset** WAV recordings for ES2002d, ES2008d, and ES2014d. Audio is downloaded from the official AMI mirror. Speaker timings are obtained as RTTM files from the AMI diarisation setup repository [3].

Each WAV file contains a single-channel waveform sampled at 16 kHz. Each parsed RTTM row contributes the meeting identifier, speech start time, duration, end time, and speaker label. A meeting-qualified label such as ES2002d/FEE005 is used so that every identity is unique across the selected meetings.

Table 1: Selected AMI recordings and annotations.

Meeting	Duration (min)	Audio (MB)	RTTM segments	Speakers
ES2002d	43.74	80.08	774	4
ES2008d	43.76	80.13	757	4
ES2014d	48.52	88.85	695	4
Total	136.02	249.06	2226	12

2.2 Enrollment and evaluation split

For each speaker, the code finds RTTM segments that (i) last from 0.8 to 10 seconds, (ii) have no temporal overlap with another annotated speaker, and (iii) end before 2100 seconds. Candidates are sorted by start time, and the first five are selected. This deterministic rule produces 60 enrollment segments, five per speaker. Their durations range from 0.82 to 9.92 seconds. The latest selected enrollment segment ends at 1826.23 seconds.

The future evaluation material is the interval from 2100 to 2400 seconds in each meeting. Thus, enrollment and evaluation are separated by at least 273.77 seconds. They are temporally disjoint but come from the same meeting sessions; this is a closed-set, session-matched design rather than a cross-session speaker-recognition test.

Table 2: RTTM reference composition of the future evaluation excerpts. Overlap is a subset of speech, not an additional duration.

Meeting	Excerpt (s)	Speech (s)	Silence (s)	Overlap (s)
ES2002d	300	264.79	35.21	50.79
ES2008d	300	238.45	61.55	21.86
ES2014d	300	139.14	160.86	10.24
Total	900	642.38	257.62	82.89

This selection yields 15 minutes of continuous evaluation audio with all four meeting participants present in each excerpt.

3 Reproducible Baseline Setup

3.1 Models and serial pipeline

The pretrained Silero VAD model, version 6.2.1, is used with its default inference settings to return sample-level speech boundaries [7]. The speaker component is `speechbrain/spkrec-ecapa-voxceleb`, an ECAPA-TDNN model pretrained on VoxCeleb data and distributed through SpeechBrain [4, 5, 6]. ECAPA-TDNN maps a variable-length waveform to a fixed-dimensional representation; this implementation produces a 192-dimensional embedding.

The planned Phase 2 pipeline is:

continuous audio $\longrightarrow$ Silero VAD $\longrightarrow$ detected intervals $\longrightarrow$ ECAPA-TDNN embeddings $\longrightarrow$ speaker predictions.

For an enrolled speaker k , five unit-normalised embeddings are averaged and the result is normalised again to form centroid $\mathbf{c}_k$. A detected interval is embedded as $\mathbf{e}$ and compared with every enrolled centroid using cosine similarity,

$$s(\mathbf{e}, \mathbf{c}_k) = \frac{\mathbf{e}^\top \mathbf{c}_k}{\|\mathbf{e}\|_2 \|\mathbf{c}_k\|_2}. \quad (1)$$

Because both vectors are unit-normalised, the implementation uses their dot product. The centroid with the highest score is the top-1 prediction. All 12 centroids are considered, not only the four speakers from the current meeting.

As an exploratory overlap heuristic, Phase 2 will also inspect the margin between the two highest similarity scores. A sufficiently small margin will return both speakers as possible candidates; otherwise only the highest-scoring speaker will be returned. The threshold value is deliberately not fixed in Phase 1. It will be selected and documented before final evaluation and will not be changed in response to test metrics. This heuristic is not presented as a reliable overlap detector because ECAPA-TDNN produces one embedding and was not designed as a multi-speaker classifier.

3.2 Environment and execution

The notebook runs in Google Colab on CPU. Its installation cell is:

```
%pip install -q speechbrain==1.1.0 silero-vad==6.2.1 soundfile==0.14.0
```

The tested environment used PyTorch 2.11.0+cpu and TorchAudio 2.11.0+cpu. The notebook downloads all three audio and RTTM files, parses annotations, selects enrollment segments deterministically, loads the pretrained models, and constructs the centroids. Only requested audio intervals are read from disk, avoiding the need to load all recordings into memory. Since the baseline performs deterministic inference and uses no random sampling, no experiment seed is required in Phase 1.

3.3 Phase 1 checks

The two components were checked independently to confirm that the baseline runs before Phase 2:

- A clean 3.11-second segment not used for enrollment had reference speaker **ES2002d/FEE005**. The top-1 prediction was correct, with cosine similarity 0.7257; the next-highest score was 0.2303. This single example is not reported as accuracy.
- Silero VAD processed a separate 30-second excerpt from seconds 2000–2030 of **ES2002d**, outside the evaluation interval. It returned nine speech intervals totalling 26.35 seconds. No VAD metric or parameter tuning was performed.

4 Phase 2 Evaluation Plan

4.1 Metric 1: Frame-Level VAD F1

This metric evaluates only the timing of speech activity. Each 300-second evaluation excerpt will be divided into non-overlapping 10-ms frames.

A reference frame will be labelled as speech if it overlaps any annotated speech interval and as non-speech otherwise. Similarly, a predicted frame will be labelled as speech if it overlaps any interval detected by Silero VAD. With speech as the positive class,

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad (2)$$

$$\text{F1} = \frac{2 \text{Precision Recall}}{\text{Precision} + \text{Recall}}. \quad (3)$$

Here, TP represents frames labelled as speech by both RTTM and Silero VAD, FP represents reference-silence frames incorrectly detected as speech, and FN represents reference-speech frames missed by Silero VAD. This metric measures the agreement between the reference and predicted speech regions; it does not evaluate which speaker is active.

4.2 Metric 2: Top-1 Speaker-Identification Accuracy

Speaker-identification accuracy will be evaluated on the actual speech intervals detected by Silero VAD. For each VAD-detected interval, the evaluation code will find every unique RTTM speaker whose annotated speech overlaps that interval by a non-zero duration.

Each detected interval will then be assigned to one of three groups:

- If the interval does not overlap any RTTM speaker, it is considered a VAD false alarm and is not included in speaker-identification accuracy.
- If the interval overlaps exactly one unique RTTM speaker, it is considered an eligible single-speaker interval.
- If the interval overlaps two or more RTTM speakers, it is treated as an overlapping or mixed-speaker interval and analysed separately.

An interval may be classified as mixed-speaker either because two speakers talk simultaneously or because one VAD interval combines consecutive turns from different speakers.

For every eligible single-speaker interval, the complete waveform produced by the VAD boundaries will be passed to ECAPA-TDNN. Its embedding will be compared with all 12 enrolled speaker centroids, and the centroid with the

highest cosine-similarity score will be the top-1 prediction. The prediction is correct when this speaker matches the only RTTM speaker associated with the interval. The metric will be calculated as

$$\text{Speaker Accuracy} = \frac{N_{\text{correct}}}{N_{\text{eligible}}}, \quad (4)$$

where N_{correct} is the number of eligible intervals with a correct top-1 speaker prediction, and N_{eligible} is the total number of eligible single-speaker intervals.

Intervals containing no reference speaker are already penalised through the VAD F1 score, while intervals containing multiple speakers are reserved for the separate overlap analysis.

5 Simple Analysis Plan

Phase 2 will inspect the following three categories or slices:

1. **VAD false alarms and missed speech.** Representative false-positive and false-negative regions will be inspected, especially around silence, breaths, hesitations, and low-energy speech.
2. **Short versus long detected intervals.** Eligible single-speaker intervals will be split at their median duration, and top-1 speaker accuracy will be compared between the two groups.
3. **Overlapping or mixed-speaker intervals.** Detected intervals associated with two or more RTTM speakers will be inspected separately. Simultaneous overlap will be distinguished from cases in which one VAD interval merges consecutive speaker turns. For intervals containing exactly two simultaneously active speakers, the analysis will examine whether the similarity-margin heuristic includes both reference speakers among its two candidates and will describe representative successes and failures.

Likely limitations include VAD boundary errors, cross-talk contamination of a single ECAPA embedding, dominance of one voice during overlap, and confusion on short or acoustically similar voices. The analysis will use small summaries and representative examples rather than advanced statistical testing.

6 Phase 2 Evaluation Execution

6.1 Serial inference and final configuration

The Phase 1 plan was executed without training or fine-tuning. RTTM annotations were used before inference to select the enrollment segments; for the evaluation excerpts, they were used only after inference to evaluate predictions. Silero produced 100, 91, and 65 intervals for ES2002d, ES2008d, and ES2014d, respectively, for a total of 256 detected intervals. The overlap-heuristic threshold was fixed at 0.05 before final evaluation. If the margin is below 0.05, the heuristic returns the top two speakers; otherwise it returns only the top speaker.

6.2 Reference grouping results

For Phase 2 analysis, the broad mixed-speaker group is divided further. Pairwise RTTM timing distinguishes true `simultaneous_overlap` from `consecutive_mixed_speakers`, where consecutive turns were merged by one VAD interval but were never simultaneous. Only single-speaker intervals enter top-1 accuracy. Consecutive mixed turns are also excluded from the overlap experiment.

7 Results

7.1 Voice-activity detection

Table 4 reports frame-level results. The pooled calculation concatenates all 90,000 frames before computing the metrics.

Table 3: Reference grouping of the 256 VAD-detected intervals. The exactly-two column is a subset of simultaneous overlap.

Meeting	False alarm	Single	Simultaneous	Consecutive	Exactly two
ES2002d	1	56	41	2	29
ES2008d	1	56	30	4	21
ES2014d	2	48	14	1	9
Total	4	160	85	7	59

Table 4: Frame-level VAD results on 10-ms frames.

Meeting	TP	FP	FN	TN	Precision	Recall	F1
ES2002d	22554	416	3959	3071	0.9819	0.8507	0.9116
ES2008d	21694	1246	2194	4866	0.9457	0.9082	0.9265
ES2014d	10558	587	3396	15459	0.9473	0.7566	0.8413
Overall	54806	2249	9549	23396	0.9606	0.8516	0.9028

The overall F1 is 0.9028. Precision (0.9606) is higher than recall (0.8516): relative to RTTM, the model creates few false-speech frames but misses more reference-speech frames. Performance also varies by meeting. **ES2008d** has the highest F1 (0.9265), while **ES2014d** has the lowest recall (0.7566) and F1 (0.8413).

7.2 Top-1 speaker identification

Table 5: Top-1 speaker identification on eligible single-speaker VAD intervals.

Meeting	Eligible intervals	Correct	Accuracy
ES2002d	56	54	0.9643
ES2008d	56	47	0.8393
ES2014d	48	45	0.9375
Overall	160	146	0.9125

The baseline correctly identifies 146 of 160 eligible intervals, giving overall top-1 accuracy of 0.9125. This result is restricted to VAD intervals associated with exactly one RTTM speaker. It does not measure false alarms, simultaneous overlap, consecutive mixed turns, unknown-speaker rejection, or complete diarisation quality.

8 Simple Breakdown and Error Analysis

8.1 Short versus long detected intervals

The 160 eligible intervals are split at their overall median duration of 1.196 seconds. Durations at or below the median form the short group; longer durations form the long group.

Short-interval accuracy is 12.5 percentage points below long-interval accuracy, and 12 of the 14 speaker mistakes occur in the short group. A likely explanation is that a short waveform provides less speaker-specific acoustic evidence for a stable embedding, although this descriptive split alone does not prove causation. The demo includes a representative 0.38-second error: the reference is **ES2002d/MEE008**, while top-1 is **ES2008d/MEE031**; the true identity appears second with a small margin of 0.0217.

8.2 Exploratory overlap results

The heuristic is evaluated only on 59 detected intervals containing simultaneous speech from exactly two total reference speakers. It returns two candidates for 6 of these intervals and only one candidate for 53. Both returned

Table 6: Speaker accuracy by duration group.

Group	Count	Mean duration (s)	Correct	Incorrect	Accuracy
Short	80	0.720	68	12	0.8500
Long	80	2.366	78	2	0.9750

candidates match the reference pair in just one case, an exploratory exact-pair success rate of $1/59 = 0.0169$. Among the 160 true single-speaker intervals, it unnecessarily returns two candidates for 28 and one candidate for 132.

Thus, a small top-two margin is not a useful overlap indicator in this experiment. The heuristic activates on 10.2% of the selected overlap intervals but on 17.5% of single-speaker intervals. This negative result supports the original limitation: one ECAPA embedding from mixed speech is not designed to preserve two speaker identities.

Table 7: Representative RTTM-confirmed two-speaker overlap cases. Times are local to the 300-second evaluation excerpt.

Case	Meeting, time (s)	Reference speakers	Returned candidates	Margin
Exact pair	ES2008d, 225.70–230.49	FEE029, FEE030	FEE030, FEE029	0.0247
Wrong second candidate	ES2002d, 74.24–75.81	FEE005, MEE008	MEE008, ES2008d/FEE030	0.0060
Overlap missed	ES2002d, 5.12–7.13	MEE006, MEE007	MEE006 only	0.2103

The first example is the only exact-pair success; candidate order is irrelevant. In the second, the top prediction is one true speaker but the additional candidate belongs to another meeting, showing that a small margin can represent model uncertainty rather than overlap. In the third, one true speaker dominates the similarity scores and the second speaker is not returned.

8.3 Manual VAD error inspection

Contiguous false-positive and false-negative frames are combined into error intervals. The two longest intervals of each type are played with 0.5 seconds of surrounding context. Table 8 records the student’s listening observations.

Table 8: Representative VAD errors selected by duration. Times are local to the evaluation excerpt.

Type	Meeting	Time (s)	Duration (s)	Manual observation
False positive	ES2008d	224.29–225.54	1.25	Laughter followed by a short “ah” vocalisation.
False positive	ES2008d	238.80–239.78	0.98	Apparent continuation of the laughter with faint speech-like sounds.
False negative	ES2014d	164.23–167.02	2.79	Very low-pitched voice recorded close to the microphone.
False negative	ES2014d	184.21–186.86	2.65	Similar close-microphone, low-pitched voice, apparently the same speaker.

The false positives show an evaluation ambiguity: laughter and short vocalisations are human voice activity that Silero can reasonably detect, while the RTTM reference labels the corresponding frames as silence. The faint content in the second case also suggests a possible boundary or annotation mismatch. The two false negatives suggest that the acoustic characteristics of one voice or recording may be difficult for the pretrained VAD. These are four qualitative examples, not general explanations for all VAD errors.

9 Limitations and Future Work

The results are limited by the following assumptions:

- Only three meetings and 15 minutes of evaluation audio are used. Enrollment and evaluation are from different times but the same sessions, so the experiment does not test cross-session robustness.
- The speaker task is closed-set. Every eligible reference speaker is enrolled, and the system cannot reject an unknown voice.

- Top-1 accuracy excludes false alarms, simultaneous overlap, and consecutive mixed-speaker intervals. It therefore should not be interpreted as end-to-end diarisation accuracy.
- One identity is assigned to a complete VAD interval. The baseline has no speaker-change detector, speech separation, or reliable multi-speaker representation.
- RTTM is treated as the evaluation reference, but its annotation convention may exclude laughter or weak vocalisations that a general VAD detects as speech.

Two reasonable future improvements follow directly from the observed errors. First, an overlap-aware diarisation method could use shorter sliding windows, speaker-change detection, or a model explicitly designed for overlapping speakers instead of the top-two-margin heuristic. Second, Silero parameters or a meeting-domain VAD could be selected on a separate validation subset to study the precision–recall trade-off for low-pitched speech and non-lexical vocalisations. Neither improvement is implemented in the current lightweight project.

10 Lightweight Demo and Future Integration

The executed Colab notebook is the lightweight demo. After running the serial pipeline, it plays three actual VAD-detected intervals: one correct single-speaker prediction, one incorrect single-speaker prediction, and the one overlap interval for which both reference speakers were returned. For each input it shows meeting ID, local start and end times, reference speakers, top-1 prediction, candidate list, cosine score, and similarity margin.

A future learning-assistant component could accept a mono 16-kHz waveform together with an enrollment-centroid dictionary and return timestamped speaker-attributed segments. A minimal output contract is:

```
{"meeting": "ES2002d",
 "segments": [{"start": 0.51, "end": 1.69,
                  "speaker": "ES2002d/MEE006",
                  "cosine_score": 0.3588}]}
```

The cosine score is a similarity value, not a calibrated probability. These segments could be passed to automatic speech recognition to produce a diarised transcript for lecture or meeting indexing.

11 Reproducibility

The repository contains the executed Colab notebook, pinned Python requirements, this report, and a README with execution instructions. The notebook downloads the three WAV and RTTM files, selects enrollment segments with a deterministic rule, loads both pretrained models, creates centroids, runs the serial pipeline, computes the two main metrics, produces the result breakdowns, and plays the demo examples. The tested runtime uses CPU inference with NumPy 2.0.2, PyTorch 2.11.0+cpu, TorchAudio 2.11.0+cpu, SpeechBrain 1.1.0, Silero VAD 6.2.1, and SoundFile 0.14.0. No experiment seed is needed because the pipeline uses no random sampling.

References

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